HemanthVarma Dantuluri

+1-410-979-4876 | hvarmadantuluri@gmail.com | LinkedIn

SUMMARY

Data Scientist with 4+ years of experience in developing scalable data pipelines and financial models across diverse sectors. Possesses a proven track record in building dynamic dashboards and cost-effective automation workflows using SQL, Python, and AWS tools. Demonstrated expertise in capacity planning, variance analysis, and data analytics, consistently driving strategic insights and optimized resource allocation.

SKILLS

- **Programming & Data Engineering**: Python (OOP, Data Engineering), SQL (PostgreSQL, Oracle), PySpark
- **Machine Learning & AI**: Predictive Modeling, Supervised & Unsupervised Learning, Feature Engineering, Segmentation, Clustering (K-Means, DBSCAN), Regression (Linear, Logistic, Multiple), Naive Bayes, XGBoost, Random Forest, SVM, KNN
- **LLMs & GenAI**: GPT-3.5/4, Hugging Face Transformers, LangChain, RAG Pipelines, Prompt Engineering
- **Big Data & ETL**: AWS (Glue, Redshift, S3, Lambda, Athena, EMR, SageMaker, Kinesis, IAM, Step Functions), Apache Spark, Hadoop, Kafka
- **Data Processing & Orchestration**: Apache Airflow, Control-M Scheduler
- **Data Visualization & BI Tools**: Tableau, Power BI, AWS QuickSight, Looker Studio
- **DevOps & Data Integration Tools**: CI/CD (Jenkins, Git, Bitbucket), REST APIs, Informatica PowerCenter
- **Methodologies**: Agile, Software Development Life Cycle (SDLC), Work Breakdown Structure (WBS), Jira

WORK EXPERIENCE

Comptroller of Maryland — **Mar 2025 - Present**
Data Scientist — *Annapolis, MD*

- Directed data science initiatives by applying predictive analytics and AWS-based solutions to support fiscal policy decisions and data governance improvements.
- Engineered AWS Lambda functions triggered by S3 events to validate and transform incoming datasets, enhancing model readiness and ensuring 99% input data quality while leveraging SQL and Python skills.
- Performed statistical modeling and hypothesis testing for policy impact studies, delivering executive-level Tableau dashboards that reduced analysis turnaround time by 40% and supported financial modeling insights for capacity planning.
- Constructed machine learning pipelines for classification and segmentation using Python, scikit-learn, and PySpark to underpin A/B testing and NLP enhancements for data-driven decision-making.
- Formulated end-to-end reporting automation workflows via Athena, S3, and Lambda, reducing manual intervention by 60% and streamlining dashboard development for key performance metrics.

Tata Consultancy Services — **Jul 2020 - Jun 2022**
Data Scientist — *Chennai, India*

- Orchestrated customer segmentation initiatives for enterprise clients using K-Means, DBSCAN, and PCA, enabling targeted marketing strategies and improving campaign ROI by 35%.
- Developed comprehensive ETL pipelines on AWS (Glue, S3, Lambda, Step Functions, Athena) to ingest, cleanse, and transform structured and unstructured data, resulting in a 25% boost in data access speed and a 20% reduction in processing errors while supporting capacity planning requirements.
- Extracted and analyzed multi-terabyte datasets with PySpark and SQL, applying advanced data wrangling techniques in Pandas and NumPy to ensure high-quality inputs for analytical financial modeling and variance analysis.
- Designed and deployed machine learning models (Random Forest, XGBoost, LSTM, Logistic Regression) in Python and scikit-learn to predict customer lifetime value, churn risk, and product demand, driving a 15% increase in product performance and revenue optimization.
- Automated event-driven data transformations using AWS Lambda and Apache Kafka, achieving 25% lower data latency and real-time monitoring of key financial and behavioral metrics for improved decision support.
- Created interactive Tableau dashboards that integrated A/B testing insights to visualize customer clusters, churn patterns, and model performance, enabling stakeholders to identify risk exposure and strategic opportunities efficiently.
- Implemented personalized recommendation engines via collaborative filtering and deep learning techniques, reducing system response times by 35% and enhancing user engagement through data-driven insights.
- Mentored a cross-functional team of seven junior engineers in agile best practices, MLOps workflows, and scalable data pipeline design, improving delivery timelines and adherence to data governance standards.

Tata Consultancy Services — **Jul 2019 - Jul 2020**
Data Engineer — *Chennai, India*

- Engineered end-to-end ETL data pipelines to ingest, clean, validate and store over 50 million financial transactions daily, ensuring 99% data accuracy.
- Migrated on-premises Hadoop infrastructure to AWS EMR, reducing infrastructure costs by 25% and improving cluster performance.
- Developed Spark jobs for large-scale data processing, reducing ETL runtimes by 60% and accelerating data availability for analytics.

- Implemented data quality checks and monitoring using Apache Airflow, decreasing data downtime by 30%.
- Collaborated with data scientists to standardize datasets and streamline data access, accelerating model deployment by 20%.

EDUCATION

University of Maryland, Baltimore County. **Aug 2022 - May 2024**
Masters in Professional Studies, Data Science
- **GPA:** 4.0
- **Achievements:** CGPA 4.0

Jawaharlal Nehru Technological University, India **Jul 2015 - May 2019**
Bachelor Of Technology, Electronics and Communication Engineering

ACADEMIC PROJECTS

Financial News Sentiment Analysis (Python, NLP, Transformers) **Jan 2024 - Present**
- Built a pipeline using Python, Hugging Face Transformers, and Flask to analyze financial news sentiment, achieving 78% accuracy in market movement prediction.
- Deployed the model with real-time API integration and interactive dashboards, enabling users to visualize sentiment-driven trading signals.

OCT Image Classification
- Designed and implemented a U-Net CNN model using TensorFlow for OCT Image classification on retinal scans, achieving an F1-score of 92% and significantly improving detection over baseline methods.
- Researched deep learning techniques for medical image classification, optimizing model accuracy and robustness through advanced data augmentation and ensemble techniques, enabling highly accurate diagnosis of critical eye conditions.
- Collected, annotated, and preprocessed a dataset of 60,000+ medical images, increasing training data volume by 40%.
- Collaborated with radiologists to translate clinical requirements into AI solutions, resulting in faster diagnostic workflows.

CERTIFICATIONS

- AWS Certified ML Engineer Associate - 2025 (in progress).
- AWS Certified Data Engineer Associate - 2024.
- SQL Advanced certificate from Hacker rank - 2023.